What is ESD?

ESD stands for “electrostatic discharge” which is basically when there are sudden releases of built up static electricity. This can be very bad as it damages parts of an electrical circuit or component. Most of the time just adding one tool or component is not enough as there are a multitude of issues that can occur with ESD.

Not only can ESD irreparably damage electrical components but can also cause sparking which can possibly ignite gasses causing fires and other things

What is ESD protection?

ESD protection helps guard sensitive electrical components and devices against the potential damaging effects of electrostatic discharge. No matter what you're doing in circuitry you will need to be wary of the potential effects and events of ESD. While some of the time ESD doesn’t do too much to your device it can also get to extreme effects such as affecting the reliance and performance of your device, minor damage to components, melting or burnout and even making the entire system fail.

What causes ESD to occur?

ESD can occur when two things of different electrostatic charge come into contact with each other. This is exactly what happens when you're on a trampoline for example and when you touch the metal frame of it you get a little shock. On a larger scale to this; it is what happens when lightning strikes the earth (very large and scary difference for me).

ESD symbols

There are 3 ESD symbols to look out for.

ESD susceptibility symbol

This is the most common type of symbol. It depicts a hand reaching to a triangle with a cross cutting through it. It is often highlighted in yellow orange as they are eye-catching colours.

This indicates that a device or components is likely to suffer ESD damage. It cautions to be careful and take proper ESD precautions while handling it.

ESD protective symbol

This image very much resembles the susceptibility symbol but it means a very different thing. The hand no longer has a cross through it and there is now a mostly complete circle around the old symbol

This symbol shows that the device or component is ESD protective. This means that it offers at least one ESD control feature lowering the risk of it happening. This is why it's called the ESD protective symbol.

ESD common point ground symbol

This is the final ESD symbol that you will need to look out for. This indicates a device that is properly grounded and can be used to discharge the build up of electricity harmlessly. They should be able to connect themselves safely to a grounding point before making contact minimising the effects of ESD.

ESD protection methods?

There are many ways to fight against ESD being things you can physically wear or implement into the circuit/components.

Things you can use to prevent ESD protection include:

- An ESD wristband that you can wear to continuously and safely drain static electricity build up,
- An ESD mat which provides a controlled safe path to drain static electricity from your body and electrical components,
- A ground cable which provide a controlled and low resistance path for static electricity to flow into the earth
- A grounding plug which provides a safe and controlled path for static electricity to flow to the earth. This neutralises the electrostatic path before it can damage any sensitive components
- An ESD loop Which works by providing a controlled path for static electricity to safely discharge making sure that your body, tools and any delicate parts remain at the same electrical potential energy.

You can also take some measures to control ESD which include but is (probably) not limited to: ESD grounding clamps, Particle Remover Rollers, Anti Static Wrist Straps, Ionisers, Ionisation accessories, Anti-Static Bags, ESD Shoes, ESD Grounding Kits, ESD Mat Kits, ESD Test Equipment, ESD Gloves, ESD Safe Tools and ESD Packaging Boxes. I will not individually explain each one of these preventative maintenances as there are just a whole bunch of them and I'm not up to that so if you want to find more read here: https://uk.rs-online.com/web/content/discovery/ideas-and-advice/esd-protection-guide or alternatively you can go to the bibliography and click the same link Which should be labeled as 1.

Why does the usb hub need it?

USB hubs need ESD protection because they are often exposed and have frequently handled ports that act as direct pathways for static electricity.

What is USB hub ESD protection?

A USB ESD is a semiconductor that is most commonly a Transient Voltage Suppressor (TVS) diode array or a low capacitance Zener based suppressor which shunts electrostatic discharge away from any sensitive circuitry the moment the USB is handled or plugged in.

If you touch a USB while your hands are charged it delivers a high voltage pulse. Without a proactive ESD device this pulse propagates directly into the USB PHY transceiver IC where it then vastly exceeds the maximum voltage capacity causing permanent damage which can never be fixed.

Advantages of USB ESD

1. ### High ESD withstanding:
   The best ESD devices can survive surges which are classified as extreme. They do this while keeping the clamp voltage which is basically just how much a TVS component will allow to pass through it meaning that you won't need as much voltage to work. THis means it works safely below the threshold at which most USBs will experience permanent damage.
2. ### Extremely fast response times
   TVS diodes activate in around 0.2 - 1 nanoseconds which is 2e-10 - 1e-9 seconds and when numbers have letters that's when you know it's a large number meaning they turn on extremely quickly. The conversion rate of seconds to nanoseconds from a quick google search is 10^-9 seconds which is around a billionth of a second. This is so fast in fact that it can intercept the leading edge of the static discharge before it is able to cause any damage at all!
3. ### Compact Multi-Chanel Array

Basically a single DFN package can protect the lanes simultaneously. This reduced the PCBs footprint by a massive 60-70%! This makes the routing vastly simpler which reduces the possible amount of asymmetries that introduce differential noise.

The different types of TVS

There are two different types of TVS diodes which are the Unidirectional and Bidirectional TVS diodes which are used in different situations.

Unidirectional TVS Diode: When the Unidirectional TVS is placed in the forward direction it will act as a diode, this means that it will conduct at the usual diode threshold. When placed in the reverse direction it will only conduct when the voltage is above the rated breakdown voltage. This type of TVS diode is meant to protect from signal voltages of one polarity. Similar to a zener diode it limits voltage between -0.6V and whatever the breakdown voltage is. This is the type of the TVS you use for USBs that always have a voltage of positive or 0 volts.

Bidirectional TVS Diode: This is basically two of the Unidirectional TVS back to back in series so that it will conduct in any direction if the breakdown voltage is to ever be exceeded. This is used to protect signals which could go positive or negative. For example audio. It will limit the positive breakdown voltage and negative breakdown voltage.

For USB C and A use Bidirectional TVS diodes

Parameter	**Unidirectional TVS**	**Bidirectional TVS**
Polarity	Single polarity	Dual polarity (±)
ESD on data lines	Requires careful orientation	Fits differential pairs directly
Standby leakage	Lower (typical <1 µA)	Slightly higher (<5 µA typical)
Clamping symmetry	Asymmetric (anode/cathode differ)	Symmetric — identical ±Vc
Typical use case	Power rail, single-ended I/O	USB D+/D-, USB3 SuperSpeed lanes
Risk of misorientation	Present — causes clamp failure	None — polarity-agnostic

This is a table of the comparison between the two TVS diode types from this website here: https://www.lcsc.com/blog/esd-protection-for-usb-ports-a-complete-engineering-guide/

How to add it to the usb hub

Always make sure that the TVS device is placed within 0.5-1mm of the USB connectors ground pins on the same layer as the connector.
Directly route the signal traces from the connector pads to the TVS device pads.
Any unprotected trace between the connector and TVS acts as antennas which radiate the ESD pulse toward the IC before the clamp activates.
Keep the connection of the TVS as short as possible and direct to the chassis.

Bibliography

Just the websites I used in this and the info I got.

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